

Hamdard University Bangladesh

Faculty of Science, Engineering and Technology

Midterm Examination (Spring'25)

PHY 121 Physics II (Mechanics, Electricity, Magnetism and Modern Physics)

Time: 1.5 hours

Marks: 30

Answer any 3 (Three) of the following sets of questions

- 1 (a) State Coulomb's Force law. 2 CLO1
(b) Deduce Coulomb's Law from Gauss's Law. 4 CLO2
(c) Four charges $+10^{-7}$ C, 10^{-7} C, -2.0×10^{-7} C, and $+2.0 \times 10^{-7}$ C are placed at the corners A, B, C, and D of a square of side 5 cm, respectively. Find the net force on the charge at corner D. 4 CLO3

- 2 (a) What is the physical significance of electric dipoles? 2 CLO1
(b) Show that electric field E due to a dipole at a point P, a distance 'x' along the perpendicular bisector of the joining the charges is $E = \frac{1}{4\pi\epsilon_0} \times \frac{p}{x^3}$ 4 CLO2
(c) The electric strength of air is about 3.0×10^6 N/C which means that if the electric field exceeds this value, sparking will occur. What is the largest charge a 0.50 cm. radius sphere can hold if sparking is not to occur in the air surrounding it? 3 CLO3

- 3 (a) State Gauss's Theorem. 2 CLO1
(b) Find an expression for the potential at a point due to a electric dipole and discuss the different cases. 4 CLO2
(c) A spherical drop of water carrying a charge of 3×10^{-6} coul has a potential of 500 volts at its surface (i) What is the radius of the drop? (ii) If two such drops of the same charge and radius combine to form a single spherical drop, what is the potential at the surface of the new drop formed? 4 CLO3

- 4 (a) What is a capacitor? Explain the term capacitance. 2 CLO1
(b) Derive an expression for the energy stored in the electric field of a charged capacitor. Then obtain an expression for energy density. 4 CLO2
(c) A plane-parallel plate capacitor has circular plates of radius $r = 10.0$ cm, separated by a distance $d = 1.00$ mm. (i) How much charge is stored on each plate when their electric potential difference has the value $V = 100$ V? (ii) Calculate the electric field energy density, and the energy stored in the capacitor. 4 CLO3



Department of Computer Science and Engineering
Bachelor of Science in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT125

Course Title: Linear Algebra and Complex Variables

Midterm Examination

Semester: Spring 2025

Date: 28/06/2025

Total Time: 1:30 hours

Total Marks: 30

3 × 10 = 30

Answer any three (03) of the following questions:

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

No.	Questions/Statements	CLOs	Marks
1.	(a) Define homogeneous linear system.	CLO1	[1]
	(b) Solve the following system of linear equations by row echelon form: $\begin{aligned} x + y + 2z &= 9 \\ 2x + 4y - 3z &= 1 \\ 3x + 6y - 5z &= 0 \end{aligned}$	CLO3	[3]
	(c) Solve the following homogeneous system of linear equations by using Gauss-Jordan elimination method. $\begin{aligned} 2x_1 + 2x_2 - x_3 + x_5 &= 0 \\ -x_1 - x_2 + 2x_3 - 3x_4 + x_5 &= 0 \\ x_1 + x_2 - 2x_3 - x_5 &= 0 \\ x_3 + x_4 + x_5 &= 0 \end{aligned}$	CLO3	[6]
2.	(a) Define matrix.	CLO1	[1]
	(b) Find the solution of linear system by using inverse matrix: $\begin{aligned} x_1 + 2x_2 + 3x_3 &= 5 \\ 2x_1 + 5x_2 + 3x_3 &= 3 \\ x_1 + 8x_3 &= 17 \end{aligned}$	CLO3	[4]
	(c) Find the inverse of the following matrix by using row canonical form: $A = \begin{bmatrix} 2 & 2 & 3 \\ 1 & 2 & 5 \\ 2 & 4 & 8 \end{bmatrix}$	CLO3	[5]
3.	(a) Define linear dependence and independence.	CLO1	[2]
	(b) Show that the set of vectors $\{(2, -1, 4), (3, 6, 2), (2, 10, -4)\}$ are linearly independent.	CLO3	[4]
	(c) Prove that the vectors $(1, 2, 0), (0, 5, 7)$ and $(-1, 1, 3)$ form a basis for R^3 .	CLO3	[4]
4.	(a) What is Hermitian matrix?	CLO1	[1]
	(b) Verify Cayley-Hamilton theorem of the matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$.	CLO3	[4]
	(c) Show that every square matrix can be uniquely expressed as sum of a symmetric matrix and a skew-symmetric matrix.	CLO3	[5]
5.	(a) Find the rank(T), Im(T), nullity and Ker(T) of the following linear transformations: $T: R^3 \rightarrow R^3$ and $T(x, y, z) = (x + 2y - z, y - x, x - z)$.	CLO3	[4]
	(b) Find the Eigenvalues and Eigenvectors of the following matrix: $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -2 & 0 \\ 0 & -5 & 2 \end{bmatrix}$	CLO3	[6]



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science and Engineering
Department of Computer Science and Engineering
BSC(Honors) in CSE
ENG 127: English Skills
Mid-Term Examination
Spring 2025

2nd Semester

Time: 1 Hour 30 Minutes

June 25, 2025

Total Marks: 30

No.	Question/ Statement	CLOs	Marks
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Section A

Read the passage carefully and answer any 20 questions from 1-22

- Polar bears are in danger due to climate change, and the loss of them could have significant effects. They live in the Arctic Circle, where temperatures can be extremely cold, sometimes reaching -40°C . One reason they survive is their thick fat layer under the skin, about 11 centimeters thick. If humans had this much fat, they would be called obese and could get diseases like diabetes or heart problems. However polar bears do not have these health problems.
- In 2014, scientist Shi Ping Liu and his team compared the genes of polar bears with brown bears, which live in warmer places. They found that polar bears have a special gene called APoB, which lowers "bad" cholesterol (LDL). In humans, changes in this gene can cause heart disease. So, studying polar bears might help us learn more about heart health.
- Polar bears' genes could also help with osteoporosis, a disease where bones become weak, especially in older people. Normally, bones lose density if people do not exercise or get enough calcium. Female polar bears fast for about six months each year when they stay in snow dens to give birth and care for their cubs. During this time, they do not eat but still keep their bones strong.
- In 2008, scientists Alanda Lennox and Allen Goodship found that female polar bears make their bones denser before fasting. When they come out of the den, their bones have not weakened much. Brown bears do not have this ability and need to rebuild their bones later.
- People often think bears are not smart, but studies show polar bears are clever. A bear named GoGo in a zoo in Japan used a tree branch as a tool to get meat. Wild polar bears have also shown smart behavior, like jumping on barrels to reach high places.
- Polar bears also show emotions. They sometimes hit the ice in frustration and can form unusual friendships. One hand-raised bear named Agee became very close to her human owner and even swam with him. This is surprising because wild polar bears usually hunt humans. If polar bears disappear due to climate change, we will lose important medical knowledge and a unique, intelligent animal.

Choose the correct heading (1–6) for paragraphs A–F. CLO1 6×1=6

- Emotional behavior and unusual social relationships of polar bears
- Female polar bears' unique bone maintenance
- Polar bears' adaptation to extreme cold
- Evidence of polar bears' intelligence and problem-solving
- Genetic traits that protect polar bears' health
- Report of statistics on female polar bear

Fill in the blanks with appropriate words from the passage.

CLO1 5×1=5

7. Polar bears have about _____ centimeters of fat under their skin.
8. The gene that lowers bad cholesterol in polar bears is called APoB.
9. Female polar bears fast for around 45 months in their maternity dens.
10. Scientists discovered that female bear makes their _____ stronger before fasting.
11. A polar bear named _____ used a tree branch as a tool in a zoo.

Find True / False / Not Given from the following

CLO2 5×1=5

12. Polar bears live in very cold places where temperatures can drop to -40°C.
13. Polar bears suffer from heart disease because of their thick fat.
14. Female polar bears lose a lot of bone strength during winter fasting.
15. Polar bears are known to cry like humans when they are sad.
16. Some polar bears use tools to solve problems in zoos and in the wild.

Choose the best answers from the alternatives.

CLO2 6×1=6

17. What helps polar bears stay warm in freezing temperatures?
A) Fast running B) Long sleep C) Thick fat under their skin D) Drinking warm water
18. What happens to female polar bears before they enter the den?
A) They become weaker B) They make their bones stronger
C) They lose their fur D) They eat only plants
19. What do brown bears need to do after coming out of den?
A) Fix their bones B) Eat only fish C) Travel to warmer places D) Build new dens
20. What did the polar bear GoGo do in the zoo?
A) Opened a gate with a rope B) Used a stick to reach the meat
C) Played with a ball D) Climbed a tree to escape
21. Why do scientists study polar bear genes?
A) To find out why they grow so fast B) To understand heart disease in humans
C) To know why they sleep so long D) To learn about their hunting style
22. What shows polar bears may have feelings?
A) They growl when they are tired B) They smile at their owners
C) They play with fish D) They hit the ice when they miss a kill

Section B

23. What is skimming? Write the differences between skimming and scanning.

CLO1 5×1=5

24. Rewrite the following passage using correct punctuation.

CLO5 5×1=5

the teacher entered the room looked around and said where is everyone today i think theyre stuck in traffic replied arif without looking up from his phone well we cant wait forever lets begin the class anyway she said firmly



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

EEE 127: Electronic Devices and Circuits

Batch: 27th

Date: 18/06/2025

Mid-Term Examination

Semester: Spring 2025

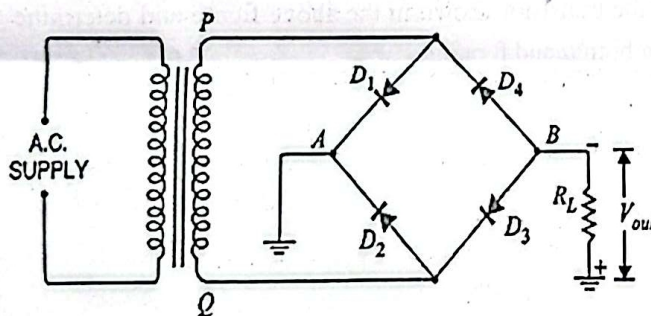
Time: 1 Hour 30 minutes

Total Marks: 30

Answer any five questions of the following.

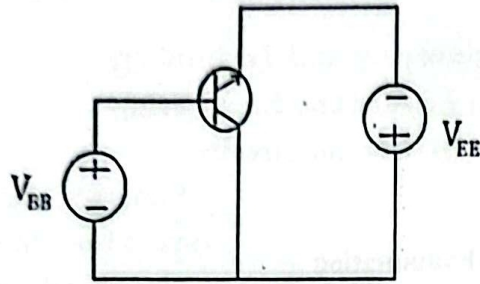
$5 \times 6 = 30$

1. (a) What is Potential Barrier? Why does it exist in a junction? CLO2 3
(b) Explain band theory with the necessary diagrams. CLO1 3
2. (a) Explain half-wave and full-wave bridge rectifiers with the necessary diagrams and signals. CLO1 3
(b) Derive the equations for Average voltage and RMS voltage in a Half-Wave Rectifier. CLO2 3
3. (a) Show two different kinds of AC and DC signals with diagram. CLO1 2



- (b) This is a step-down transformer with a 15:1 ratio. The input voltage is 220V and the frequency is 60Hz. Now calculate the DC output Voltage, Efficiency, PIV & Output Frequency for this transformer. CLO3 4
4. (a) Find the value of β if (i) $\alpha = 0.9$ (ii) $\alpha = 0.98$ (iii) $\alpha = 0.99$. CLO2 2
(b) Analyze three types of transistor configuration with appropriate figures. CLO2 4
5. (a) Explain the limitations of the p-n junction. CLO1 3
(b) Explain the I-V Curve of p-n junction for both Forward and Reverse Bias. CLO2 3

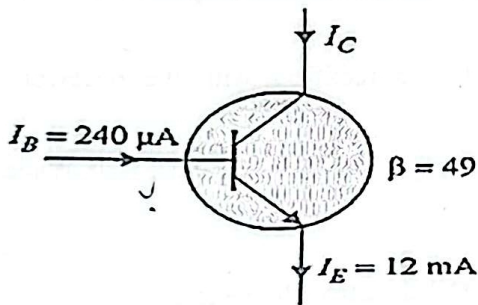
6.



(a) What type of transistor configuration does the above circuit represent? Show the current direction of the circuit. CLO2 3

(b) Calculate Emitter Current I_E in a transistor for which $\beta = 50$ and $I_B = 20 \mu A$. CLO3 3

a) Write two properties of Semiconductor. CLO1 2



(b) Find the α rating of the transistor shown in the above figure and determine the value of I_C using both α and β rating. CLO3 4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 121 : Structured Programming Language

Batch: 27th

Date: 21/06/2025

Midterm Examination

Semester: Spring 2025

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions.

5 × 6 = 30

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|---|------|-------|
| 1. (a) Define the followings — | CLO1 | 3×1=3 |
| i. Programming Language | | |
| ii. Syntax | | |
| iii. Structured Programming | | |
| (b) Differentiate between Compiler and Interpreter. | CLO3 | 3 |
| 2. (a) State the key steps in Problem Solving with Computers. | CLO2 | 2 |
| (b) Write the algorithm and draw the corresponding flowchart to determine the largest number among three given numbers (num1, num2 and num3). | CLO2 | 4 |
| 3. (a) What is the role of #include directive in C programming? | CLO4 | 2 |
| (b) Identify and Correct errors in the following programs — | CLO5 | 4×1=4 |
| i. #include <stdio.h>
int main() {
int a, b;
result = a + b;
printf("%d", result);
return 0;
} | | |
| ii. #include <stdio.h>
INT MAIN()
{
Printf("Welcome!!");
return 0;
} | | |
| iii. #include <stdio.h>
int main() {
char a, b;
printf("Enter the characters\n");
scanf("%d %d", a, b);
} | | |
| iv. #include <stdio.h>
void main() {
printf("Hello");
return 0;
} | | |



4. (a) State the differences between keywords and identifiers. CLO3 2
(b) Classify the following variable names as valid or invalid. For the invalid ones state the reason. CLO4 4×0.5=2
- `%variable`
 - `1stNumber`
 - `total_amount`
 - `return`
- (c) Describe the storage size and precision difference between float and double data types. CLO3 2
5. (a) Predict the output for the following code snippets and explain the type conversion: CLO4 2×1.5=3
- `int a = 5;`
`float b = 2.0;`
`float result = a / b;`
`printf("%f", result);`
 - `int a = 9;`
`int b = 2;`
`float result = (float)a / b;`
`printf("%f", result);`
- (b) Show the output for the following code snippet — CLO4 3
- ```
#include <stdio.h>
int main() {
 int x, y, z, w;
 x = 5;
 y = 10;
 z = x++ + --y;
 printf("x = %d y = %d z = %d\n", x, y, z);

 w = ++x * y--;
 printf("x = %d y = %d w = %d\n", x, y, w);

 printf("z += w = %d\n", z += w);
 printf("%d\n", (x > y) ? x - y : y - x);

 return 0;
}
```
6. (a) Determine the output for the following printf lines: CLO4 2
- ```
float num = 123.4567;
printf("%f\n", num);
printf("%.2f\n", num);
printf("%.10.3f\n", num);
printf("%.6.1f\n", num);
```
- (b) Write the following programs in C — CLO4 2×2=4
- Program to check if a given year is leap year or not.
 - Program to swap the values of two integer variables without using any third variable.



7. Write a C program that implements a simple calculator using a switch-case statement. CLO4

6

Requirements:

- i. Input two numbers (num1 and num2) using scanf.
- ii. Input an operation (+, -, *, or /).
- iii. Select the operation using a switch-case and perform the operation.
- iv. Display the result.
- v. Handle division by zero.
- vi. Handle false input.